Decision Analytics Journal 9 (2023) 100351

Contents lists available at ScienceDirect

Decision Analytics Journal

journal homepage: www.elsevier.com/locate/dajour

Picture fuzzy entropy: A novel measure for managing uncertainty in multi-criteria decision-making

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ARTICLE INFO

Keywords:
Picture fuzzy sets
Entropy
Multi-criteria decision-making
Fuzzy measures
Uncertainty

ABSTRACT

Entropy plays a vital role within fuzzy set theory as it quantifies the level of uncertainty present in the data. This study proposes a novel entropy measure tailored for Picture Fuzzy Sets (PIFS). This entropy measure is formulated based on the axioms developed for PIFS entropy, ensuring its suitability for such sets. We use a validation process to ascertain the effectiveness of this proposed measure. Furthermore, a compelling example is presented to demonstrate the practical application of entropy measures in real-world scenarios involving Multi-Criteria Decision-Making (MCDM). The findings underscore the significance of entropy measures in proficiently managing uncertainty and accommodating diversity. The proposed entropy measure is an invaluable tool for making well-informed decisions when confronted with intricate decision-making situations

1. Introduction

Decisions are a crucial part of our everyday lives. In various domains, such as healthcare, engineering, hospitality, education, and the environment, among others, decisions are made to tackle a wide range of challenges. It is essential to consider specific criteria or attributes that define the characteristics of each alternative and assist us in identifying the best option. One approach to address this issue is to focus on a single criterion and choose the alternative that aligns with that criterion. For example, one might opt for the least expensive mobile phone available in the market, considering only the cost criterion, while disregarding other factors like storage space, appearance, camera quality, etc. However, in real-life scenarios, complex decisions are not based on a single criterion; multiple criteria are taken into account before arriving at a conclusion. Therefore, in a strategic setting, Multi-Criteria Decision Making (MCDM) comes into play. MCDM allows us to select the best alternative by efficiently considering multiple criteria. Over the past few years, significant advancements have been made in MCDM techniques, which have proven effective in addressing a variety of problems. Abanda et al. [1] conducted a systematic review to examine the application of MCDM in assessing nationally determined contribution projects.

It is not always possible to quantify each attribute, necessitating a more descriptive approach. Therefore, decision-makers may utilize linguistic terms and phrases to define the characteristics of the alternatives. For example, the appearance of a product can be categorized as "good", "excellent", or "bad". However, these linguistic terms can have vague or ambiguous meanings. To address this issue, Zadeh's Fuzzy Set Theory (FST) [2], introduced in 1965, proves invaluable by employing fuzzy numbers to account for the imprecise nature of the data provided by the decision-maker and delivering a reliable solution [3]. FST is a mathematical framework that deals with uncertainty and imprecision in data by representing membership degrees. Essentially, fuzzy numbers offer information about the degree to which an element belongs to a set and the extent of that belongingness [4]. It not only quantifies qualitative data but also accommodates quantitative data to ensure uniformity.

Due to its utility, many researchers have integrated the concept of fuzzy sets various MCDM techniques to solve several problems [5,6]. But the classical FS theory does not include the uncertainty one can have while deciding. At times, the decision maker may face uncertainty regarding the inclusion or exclusion of an element from a set. In such scenarios, Atanassov's [7] Intuitionistic Fuzzy Set (IFS) comes handy, which contains a membership degree (γ), a non-membership degree (η), and hesitancy degree (ξ) and satisfies the condition $\gamma + \eta + \xi = 1$. Many researchers saw potential in this theory and the reliability it is offering. Therefore, several authors implemented it in various applications like, image processing [8,9], pattern recognition [10,11], decision making [12,13], etc.

Due to its utility, many researchers have integrated the concept of fuzzy sets into various MCDM techniques to solve numerous problems [5,6]. However, classical FST does not encompass the uncertainty that decision-makers can face. There are instances when decision-makers encounter uncertainty about whether to include or exclude an element from a set. In such scenarios, Atanassov's Intuitionistic Fuzzy Set (IFS) becomes valuable, as it incorporates a membership degree (γ),

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https://doi.org/10.1016/j.dajour.2023.100351
Received 14 August 2023; Received in revised form 15 October 2023; Accepted 27 October 2023
Available online 31 October 2023
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a non-membership degree (η), and hesitancy degree (ξ) and satisfying the condition $\gamma + \eta + \xi = 1$ [7]. This theory has gained recognition for its potential and reliability. Consequently, many researchers have applied it in various applications such as image processing [8,9], pattern recognition [10,11], decision making [12,13], and more. However, both Fuzzy and Intuitionistic Fuzzy Sets have limitations in handling all types of uncertainty that can arise. Different perspectives may lead to varied responses, including "yes", "no", "abstain", and "refusal". This realization has given rise to another type of fuzzy set known as Picture Fuzzy Set (PIFS) [14].

Several scholars have explored the applicability of FST in a variety of research areas. The concept is extensively utilized in MCDM since decision-making often involves a considerable degree of uncertainty [15,16]. PIFS is an enhanced version that encompasses a wider range of potential answers. It accommodates four types of responses: 'yes,' 'no,' 'abstain,' and 'refusal.' This can be illustrated more effectively using the example of voting. In a national election, citizens hold specific opinions about various political parties and leaders, leading to four distinct choices: voting for a party, voting against a party, remaining neutral, or refusing to participate in the process. As a result, PIFS introduces four types of degrees positive membership (γ), negative membership (η), neutral membership (θ), and refusal membership (ξ), satisfying the condition $\gamma+\theta+\eta+\xi = 1$. It has made the decision-making process more accurate and reliable as compare to other types of FS like, Intuitionistic Fuzzy Set, Pythagorean Fuzzy Set, etc. Due to its enhanced precision and greater realm, several researchers are finding potential in it and have developed various operations, measures, and applications of the same. Several MCDM problems have also been in Picture Fuzzy Environment [17,18]. Some operational laws were introduced by Zhang et al. [19] and aggregation operators for MCDM was introduced by Wei [20] in PIFS environment. In 2016, Wei [21] proposed a new cross entropy and applied it in MCDM in PIFS. Two years later, in 2018, Wang et al. [22] solved the problem of risk calculation that is associated with construction and came up with integrating VIekriterijumsko KOmpromisno Rangiranje (VIKOR) and PIFS. In 2020, Joshi [23] introduced a new PIFS information measure that used R-norm conceptualization and VIKOR method. Similarly, several integrations of fuzzy and MCDM techniques were also developed like, fuzzy Analytic Hierarchy Process (AHP)- Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) [24,25], AHP- VIKOR [26], etc.

Entropy measure plays an important role in fuzzy set [27]. It was first proposed by Shannon [28] in 1948 which provides the idea of measuring the uncertainty in the information. It is a decision-making tool that has reached out to many scholars and researchers and proven its ability to provide a comprehensive analysis of complex systems. Burillo and Bustince [29] introduced the concept of intuitionistic fuzzy entropy based on the unique properties of intuitionistic fuzzy sets, which has captured the interest of numerous researchers. Since then, it has become an emerging tool with great potential to measure the information of IFS [30] and PIFS and has been applied to various fields such as artificial intelligence, risk assessment [31], machine learning [32] and decision-making. For instance, in 2012, R-norm IF information measure was introduced by Bajaj et al. [33].

This concept can be implemented to determine the weights of criteria and decision-makers in MCDM problems. These weights signify the relative importance of each decision-maker and criterion in the decision-making process. Primarily, there are two approaches for assigning weights: the subjective approach and the objective approach. In the subjective approach, the allocation of weights to each criterion and decision-maker relies on various factors based on human intelligence. For example, the weight of a decision-maker can be determined based on their experience, skills, area of expertise, and other relevant qualifications. Regarding the weights of criteria, these can be assigned based on the subjective opinions, knowledge, and rational judgment of those involved in the decision-making process. On the other hand, the objective approach involves determining weights using mathematical formulas derived from data provided in the decision matrix. This objective method is free from bias and relies on mathematical logic, making it a superior choice for weight determination compared to the subjective approach. One widely used method for weight assessment is the entropy method, which quantifies the degree of uncertainty or diversity associated with the criteria. This allows decision-makers to gauge the level of information contained in each criterion. The same approach can be applied to calculate the weights of decision-makers.

Arya and Kumar [34] have established certain axioms that are employed to validate a measure as an entropy measure. These axioms serve as the basis for proving the validity of our developed method as an entropy measure.

In this paper, we employ the TOPSIS, a MCDM method, to address MCDM challenges within the picture fuzzy environment. TOPSIS is considered an enhanced and versatile MCDM method, surpassing other existing techniques. It was initially introduced by Hwang and Yoon in 1981 to measure the similarity between the best possible solution and the available alternatives. In this context, the 'ideal solution' signifies the optimal alternative, while the 'worst solution' represents the opposite end of the spectrum. TOPSIS determines the ranking of alternatives by calculating the distances between each alternative and the ideal solution. TOPSIS has found applications in various fields. For instance, in engineering, it has been used to evaluate the performance of different manufacturing systems, select suppliers, and assess the sustainability of energy systems. Rubayet Karim et al. [35] had proposed the selection of the optimal machine which is best in terms of both for quick production and revenue generation in the modern economy. In management, TOPSIS has been used for project selection [36], performance evaluation [37], and risk assessment [38,39]

The significance of weighting and its extensive application across various fields served as a compelling motivation to delve deeper into the topic and develop an entropy measure for PIFS. It is indeed fascinating to observe that a concept as inherently uncertain and influenced by psychological factors as 'the importance of criteria and decision-makers in the decision-making process' can be quantified with a robust mathematical approach, substantially reducing the level of ambiguity in the information. Furthermore, it is noteworthy that despite the high relevance of entropy measures in the context of PIFS, there is limited research conducted on this subject. This highlights the untapped potential for further exploration and development in this area.

The primary objective of this study is to propose and validate a novel entropy measure specifically designed for PIFS. This entropy measure will be developed based on axioms tailored to the unique characteristics of PIFS, ensuring its suitability for handling uncertainty in this extended fuzzy set framework.

To achieve this overarching goal, we will address the following research questions:

RQ1: How can we formulate a robust entropy measure for PIFS that accurately quantifies uncertainty within these sets?

RQ2: How does the proposed PIFS entropy measure perform in practical applications, particularly in the context of MCDM?

By addressing these research questions, we aim to contribute to the understanding of uncertainty modeling in decision-making, particularly in scenarios involving picture fuzzy sets. Additionally, we seek to provide decision-makers with a valuable tool for making well-informed decisions when faced with intricate and uncertain decision-making situations.

2. Preliminary

Understanding the fundamental concepts of entropy and PIFS is essential for the development of innovative measures and techniques for managing uncertainty and diversity effectively. These concepts form the basis for the subsequent discussion and analysis in this research paper, which proposes and validates a new entropy measure designed specifically for PIFS.

Definition 1 (*Fuzzy Set [2]*). Consider a Fuzzy Set P on $U = \{a_1, a_2, a_3 \dots a_n\}$ is given as:

$$P = \{(a_i, \gamma_P(a_i)) : a_i \in U\} \tag{1}$$

where $\gamma_P : U$ ™ $[0, 1]$

It has only two degrees, i.e., the membership degree $\gamma_P(a_i)$ which represents the belongingness of the element in the set and the non-membership degree $(1-\gamma_P(a_i))$ which represents the non-belongingness. An example of Fuzzy Number (FN) is as follows: $(a_1, 0.7)$ means that, 0.70 is the grade of its belongingness to the set. To be precise, it suggests only two types of answers, "yes" or "no". To include an option of "not sure" the hesitancy degree was introduced by Atanassov and it was termed as Intuitionistic Fuzzy Set (IFS).

Definition 2 (*Intuitionistic Fuzzy Set [7]*). Consider an IFS P on $U = \{a_1, a_2, a_3 \dots a_n\}$ is given as:

$$P = \{(a_i, \gamma_P\ a_i\ , \eta_P\ a_i\ ) : a_i \in U\} \tag{2}$$

where $\gamma_P, \eta_P \in [0, 1]$ are membership and non-membership functions of P respectively. It satisfies the condition:

$$0\ \mathrm{f}\ \gamma_P\ a_i\ + \eta_P\ a_i\ \ \mathrm{f}\ 1 \tag{3}$$

Atanassov [7] defined a function $\xi_P : U$ ™ $[0, 1]$ as:

$$\xi_P(a_i) = 1 - \gamma_P\ a_i\ - \eta_P\ a_i \tag{4}$$

$\xi_P(a_i) \in [0, 1]$ denotes the hesitancy degree or intuitionistic index which represents the uncertainty or indeterminacy. From the definition, it is clear that if $\xi_P\ a_i\ = 0$ then IFS become FS.

IFS has three degrees, membership degree (γ_P), non-membership degree (η_P), and hesitancy degree (ξ_P). An example of IFN (Intuitionistic Fuzzy Number) is as follows: $(a_1, 0.6, 0.2)$ means that, 0.60 is the degree of belongingness, 0.20 is the degree of non-belongingness and 0.20 is the degree of uncertainty.

But human judgment is not restricted to only yes/no/not sure kind of answers. Sometimes there can be refusal factor, which should also be included. Increasing the domain of answers increases the reliability of the solution that one gets after solving the MCDM problem. Therefore, another kind of FS called as PIFS (Picture Fuzzy Set) was introduced.

Definition 3 (*Picture Fuzzy Set*). Consider a PIFS P on $\mathrm{U} = \{\mathrm{a}_1, \mathrm{a}_2, \mathrm{a}_3 \dots \mathrm{a}_n\}$ is given as:

$$\mathrm{P} = \{(\mathrm{a}_i, \gamma_P(\mathrm{a}_i), \theta_P(\mathrm{a}_i), \eta_P(\mathrm{a}_i)) : \mathrm{a}_i \in \mathrm{U}\} \tag{5}$$

where $\gamma_P, \theta_P, \eta_P : \mathrm{U}$ ™ $[0, 1]$ denotes the membership, neutral and non-membership function of P respectively. Also, $\gamma_P(\mathrm{a}_i) \in [0, 1]$ denotes the +ve degree, $\theta_P(\mathrm{a}_i) \in [0, 1]$ denotes the neutral degree and $\eta_P(\mathrm{a}_i) \in [0, 1]$ denotes the −ve degree of $\mathrm{a}_i \in \mathrm{U}$. It satisfies the condition:

$$0\ \mathrm{f}\ \gamma_P\ \mathrm{a}_i\ + \theta_P\ \mathrm{a}_i\ + \eta_P(\mathrm{a}_i)\ \mathrm{f}\ 1 \tag{6}$$

Cuong [14] defined a function $\xi_P : \mathrm{U}$ ™ $[0, 1]$ as:

$$\xi_P\ \mathrm{a}_i\ = 1 - \gamma_P\ \mathrm{a}_i\ - \theta_P\ \mathrm{a}_i\ - \eta_P(\mathrm{a}_i) \tag{7}$$

$\xi_P\ a_i\ \in [0, 1]$ represents the refusal degree. An example of PIFN is as follows: $(\mathrm{a}1, 0.4, 0.3, 0.2)$ means that, $0.40, 0.30, 0.20, 0.10$ is +ve degree, neutral degree, -ve degree, and refusal degree, respectively.

Definition 4 (*Operation and Properties of PIFS [34]*). Let us consider two PIFS P and Q defined on $\mathrm{U} = \{\mathrm{a}_1, \mathrm{a}_2, \mathrm{a}_3 \dots \mathrm{a}_n\}$ } is given as $\mathrm{P} = \{(\mathrm{a}_i, \gamma_P(\mathrm{a}_i), \theta_P(\mathrm{a}_i), \eta_P(\mathrm{a}_i)) : \mathrm{a}_i \in \mathrm{U}\}$ and

$$\mathrm{Q} = \{(\mathrm{a}_i, \gamma_Q(\mathrm{a}_i), \theta_Q(\mathrm{a}_i), \eta_Q(\mathrm{a}_i)) : \mathrm{a}_i \in \mathrm{U}\}$$

Following the properties and operations that can be defined on the two PIFS:

1. $\mathrm{P} \subseteq \mathrm{Q}$ iff $\forall \mathrm{a}_i \in \mathrm{U}, \gamma_P\ \mathrm{a}_i\ \ \mathrm{f}\ \gamma_Q\ \mathrm{a}_i\ , \theta_P\ \mathrm{a}_i\ \ \mathrm{f}\ \theta_Q\ \mathrm{a}_i\ , \eta_P\ \mathrm{a}_i\ \ \mathrm{g}\ \eta_Q\ \mathrm{a}_i$
2. $\mathrm{P} = \mathrm{Q}$ iff $\forall \mathrm{a}_i \in \mathrm{U}, \mathrm{P} \subseteq \mathrm{Q}$ and $\mathrm{Q} \subseteq \mathrm{P}$
3. P^C denotes the complement of the set P, i.e., $\mathrm{P}^C = \{(\mathrm{a}_i, \eta_P(\mathrm{a}_i), \theta_P(\mathrm{a}_i), \gamma_P(\mathrm{a}_i) : \mathrm{a}_i \in \mathrm{U})\}$
4. $\mathrm{P} \cup \mathrm{Q} = \{\gamma_P\ \mathrm{a}_i\ \vee \gamma_Q\ \mathrm{a}_i\ , \theta_P\ \mathrm{a}_i\ \wedge \theta_Q\ \mathrm{a}_i\ \text{ and } \eta_P\ \mathrm{a}_i\ \wedge \eta_Q\ \mathrm{a}_i\ : \mathrm{a}_i \epsilon \mathrm{U}\}$
5. $\mathrm{P} \cap \mathrm{Q} = \{\gamma_P\ \mathrm{a}_i\ \wedge \gamma_Q\ \mathrm{a}_i\ , \theta_P\ \mathrm{a}_i\ \vee \theta_Q\ \mathrm{a}_i\ \text{ and } \eta_P\ \mathrm{a}_i\ \vee \eta_Q\ \mathrm{a}_i\ : \mathrm{a}_i \epsilon \mathrm{U}\}$

2.1. Distance measures

Distance measure was introduced by Atanassov in 1986. These measures help us to understand the similarity and differences between the PIFSs. Higher distance indicates that the PIFSs have greater difference and lower similarity. Some of the most common distance measures proposed by various researchers are listed

Definition 5 (*The Hamming Distance for PIFS [14,40]*).

$$d\,P_H\,(P, Q) = \frac{1}{3} \sum_{i=1}^{\text{É}} \left(\left| \gamma_P\ a_i\ - \gamma_Q\ a_i\ \right| + \left| \theta_P\ a_i\ - \theta_Q\ a_i\ \right| + \left| \eta_P\ a_i\ - \eta_Q\ a_i\ \right| \right) \tag{8}$$

Definition 6 (*The Normalized Euclidean Distance for PIFS*).

$$d\,P_{NE}(P, Q) = \sqrt{\frac{1}{2n} \sum_{i=1}^{\text{É}} \left(\gamma_P\ a_i\ - \gamma_Q\ a_i\ \right)^2 + \left(\theta_P\ a_i\ - \theta_Q\ a_i\ \right)^2 + \left(\eta_P\ a_i\ - \eta_Q\ a_i\ \right)^2} \tag{9}$$

Definition 7. Consider a real function $\mathrm{E_{FS}(P)}$, where P is a FS. If the Set P fulfills the following postulates, then it will be termed as FS entropy. [41]

FS1. *Sharpness:* $\mathrm{E_{FS}(P)} = 0$ is minimum iff P is crisp set.
FS2. *Maximality:* $\mathrm{E_{FS}(P)} = 1$ is maximum iff $\gamma_P\ \mathrm{a}_i\ = \eta_P\ \mathrm{a}_i\ = \frac{1}{2}, \forall \mathrm{a}_i \epsilon U$
FS3. *Resolution:* $\mathrm{E_{FS}(P)}\ \mathrm{f}\ \mathrm{E_{FS}(Q)}$, if P is crispier than Q.
FS4. *Symmetry:* $\mathrm{E_{FS}(P)} = \mathrm{E_{FS}(P^C)}$, where $\mathrm{P^C}$ is the complement of set P.

Definition 8. Consider a real function $\mathrm{E_{IFS}(P)}$, where P is an IFS, is called entropy of IFS if it satisfies the following axioms. [42]

IFS1. *Sharpness:* $\mathrm{E_{IFS}(P)} = 0$ is minimum iff P is crisp set.
IFS2. *Maximality:* $\mathrm{E_{IFS}(P)} = 1$ is maximum iff $\gamma_P\ a_i\ = \eta_P\ a_i\ = \xi_P\ a_i\ = \frac{1}{3}, \forall a_i \epsilon U$.
IFS3. *Resolution:* $\mathrm{E_{IFS}(P)}\ \mathrm{f}\ \mathrm{E_{IFS}(Q)}$, if P is crispier than Q i.e., $\gamma_P\ \mathrm{f}\ \gamma_Q$ and $\eta_P\ \mathrm{f}\ \eta_Q$ for $\max(\gamma_Q, \eta_Q)\ \mathrm{f}\ 1/3$ and $\gamma_P\ \mathrm{g}\ \gamma_Q$ and $\eta_P\ \mathrm{g}\ \eta_Q$ for $\min(\gamma_Q, \eta_Q)\ \mathrm{g}\ 1/3$.
IFS4. *Symmetry:* $\mathrm{E_{IFS}(P)} = \mathrm{E_{IFS}(P^C)}$, where $\mathrm{P^C}$ is the complement of set P.

Definition 9. Consider a real function $\mathrm{E_{PIFS}(P)}$, where P is a PIFS. If the Set P fulfills the following postulates, then it will be termed as PIFS entropy. [34]

PIFS1. *Sharpness:* $\mathrm{E_{PIFS}(P)} = 0$ is minimum iff P is crisp set.
PIFS2. *Maximality:* $\mathrm{E_{PIFS}(P)} = 1$ is maximum iff $\gamma_P\ a_i\ = \theta_P\ a_i\ = \eta_P\ a_i\ = \xi_P\ a_i\ = \frac{1}{4}, \forall a_i \epsilon U$.
PIFS3. *Resolution:* $\mathrm{E_{PIFS}(P)}\ \mathrm{f}\ \mathrm{E_{PIFS}(Q)}$, if P is crispier than Q i.e., $\gamma_P\ \mathrm{f}\ \gamma_Q, \theta_P\ \mathrm{f}\ \theta_Q$ and $\eta_P\ \mathrm{f}\ \eta_Q$ for $\max(\gamma_Q, \theta_Q, \eta_Q)\ \mathrm{f}\ 1/4$ and $\gamma_P\ \mathrm{g}\ \gamma_Q, \theta_P\ \mathrm{g}\ \theta_Q$ and $\eta_P\ \mathrm{g}\ \eta_Q$ for $\min(\gamma_Q, \theta_Q, \eta_Q)\ \mathrm{g}\ 1/4$.
PIFS4. *Symmetry:* $\mathrm{E_{PIFS}(P)} = \mathrm{E_{PIFS}(P^C)}$, where $\mathrm{P^C}$ is the complement of set P.

Definition 10. The Hessian Matrix ($H(\psi)$) of 4 variables for a function $\psi\ \tau_1, \tau_2, \tau_3, \tau_4$ is:

$$H(\psi) = \begin{bmatrix} \frac{\partial^2\psi}{\partial\tau_1^2} & \frac{\partial^2\psi}{\partial\tau_2\partial\tau_1} & \frac{\partial^2\psi}{\partial\tau_3\partial\tau_1} & \frac{\partial^2\psi}{\partial\tau_4\partial\tau_1} \\ \frac{\partial^2\psi}{\partial\tau_1\partial\tau_2} & \frac{\partial^2\psi}{\partial\tau_2^2} & \frac{\partial^2\psi}{\partial\tau_3\partial\tau_2} & \frac{\partial^2\psi}{\partial\tau_4\partial\tau_2} \\ \frac{\partial^2\psi}{\partial\tau_1\partial\tau_3} & \frac{\partial^2\psi}{\partial\tau_2\partial\tau_3} & \frac{\partial^2\psi}{\partial\tau_3^2} & \frac{\partial^2\psi}{\partial\tau_4\partial\tau_3} \\ \frac{\partial^2\psi}{\partial\tau_1\partial\tau_4} & \frac{\partial^2\psi}{\partial\tau_2\partial\tau_4} & \frac{\partial^2\psi}{\partial\tau_3\partial\tau_4} & \frac{\partial^2\psi}{\partial\tau_4^2} \end{bmatrix} \tag{10}$$

If $H(\psi)$ is positive definite than it automatically becomes strictly convex but if it turned out to be negative definite than it is said to be strictly concave.

Property 1. *The following inequality holds for any $P, Q \epsilon PIFS(U)$ such that P is crisper than Q.*

$$\left(\gamma_P(a_i) - \frac{1}{4}\right) + \left(\theta_P(a_i) - \frac{1}{4}\right) + \left(\eta_P(a_i) - \frac{1}{4}\right) + \left(\xi_P(a_i) - \frac{1}{4}\right)$$
$$\geq \left(\gamma_Q(a_i) - \frac{1}{4}\right) + \left(\theta_Q(a_i) - \frac{1}{4}\right) + \left(\eta_Q(a_i) - \frac{1}{4}\right) + \left(\xi_Q(a_i) - \frac{1}{4}\right)$$

and

$$\left(\gamma_P(a_i) - \frac{1}{4}\right)^2 + \left(\theta_P(a_i) - \frac{1}{4}\right)^2 + \left(\eta_P(a_i) - \frac{1}{4}\right)^2 + \left(\xi_P(a_i) - \frac{1}{4}\right)^2$$
$$\geq \left(\gamma_Q(a_i) - \frac{1}{4}\right)^2 + \left(\theta_Q(a_i) - \frac{1}{4}\right)^2 + \left(\eta_Q(a_i) - \frac{1}{4}\right)^2 + \left(\xi_Q(a_i) - \frac{1}{4}\right)^2 \tag{11}$$

3. Proposed entropy

The development of this entropy measure involved a careful analysis of the characteristics and properties of PIFS. By considering the degrees of membership, non-membership, and hesitation present in a PIFS, we have designed an entropy measure that captures the inherent uncertainty and vagueness associated with such sets.

Let us consider PIFS with +ve membership (γ_P), -ve membership (η_P), neutral membership (θ_P), refusal (ξ_P) for set P. Then the degree of fuzziness (E_{PIFS}) of PIFS P is given by

$$\begin{aligned} E_{PIFS}(P) = \frac{-1}{3n\ln 2}\sum_{i=1}^{n} & \gamma_P(a_i)\ln\gamma_P(a_i) + \eta_P(a_i)\ln\eta_P(a_i) \\ & + \gamma_P(a_i)\ln\gamma_P(a_i) + \theta_P(a_i)\ln\theta_P(a_i) \\ & + \gamma_P(a_i)\ln\gamma_P(a_i) + \xi_P(a_i)\ln\xi_P(a_i) \\ & + \eta_P(a_i)\ln\eta_P(a_i) + \theta_P(a_i)\ln\theta_P(a_i) \\ & + \eta_P(a_i)\ln\eta_P(a_i) + \xi_P(a_i)\ln\xi_P(a_i) \\ & + \theta_P(a_i)\ln\theta_P(a_i) + \xi_P(a_i)\ln\xi_P(a_i) \end{aligned} \tag{12}$$

The establishment of this entropy measure for PIFS relies on the formulation of a set of axioms specifically designed for PIFS entropy [34]. These axioms serve as fundamental principles that guide the development and definition of the entropy measure.

PIFS1. Sharpness:

Let P be a crisp set then we have the following conditions

$$\begin{aligned} &\text{(a)}\quad \gamma_P(a_i) = 1 \text{ and } \eta_P(a_i) = \theta_P(a_i) = \xi_P(a_i) = 0 \\ &\text{(b)}\quad \eta_P(a_i) = 1 \text{ and } \gamma_P(a_i) = \theta_P(a_i) = \xi_P(a_i) = 0 \\ &\text{(c)}\quad \theta_P(a_i) = 1 \text{ and } \gamma_P(a_i) = \eta_P(a_i) = \xi_P(a_i) = 0 \\ &\text{(d)}\quad \xi_P(a_i) = 1 \text{ and } \gamma_P(a_i) = \eta_P(a_i) = \theta_P(a_i) = 0 \end{aligned} \tag{13}$$

If condition (a) is true then,

$$\Rightarrow E_{PIFS} = \frac{-1}{3n\ln 2}\sum_{i=1}^{n}[(1)\ln(1) + (1)\ln(1) + (1)\ln(1) + (0)\ln(0) + (0)\ln(0) + (0)\ln(0)]$$

$$\Rightarrow E_{PIFS} = \frac{-1}{3n\ln 2}\sum_{i=1}^{n}[0] \quad [Since, 1\ln 1 = 0 \text{ and } 0\ln 0 = 0]$$

$$\Rightarrow E_{PIFS} = 0 \tag{14}$$

Similarly, if conditions (b), (c), and (d) is true than $E_{PIFS} = 0$ Now, Let $E_{PIFS} = 0$,

$$\begin{aligned} 0 = \frac{-1}{3n\ln 2}\sum_{i=1}^{n} & \gamma_P(a_i)\ln\gamma_P(a_i) + \eta_P(a_i)\ln\eta_P(a_i) \\ & + \gamma_P(a_i)\ln\gamma_P(a_i) + \theta_P(a_i)\ln\theta_P(a_i) \\ & + \gamma_P(a_i)\ln\gamma_P(a_i) + \xi_P(a_i)\ln\xi_P(a_i) \\ & + \eta_P(a_i)\ln\eta_P(a_i) + \theta_P(a_i)\ln\theta_P(a_i) \\ & + \eta_P(a_i)\ln\eta_P(a_i) + \xi_P(a_i)\ln\xi_P(a_i) \\ & + \theta_P(a_i)\ln\theta_P(a_i) + \xi_P(a_i)\ln\xi_P(a_i) \end{aligned} \tag{15}$$

Since, $\frac{-1}{3n\ln 2} \neq 0$. Therefore, the summation part must be 0

By the definition we know that,

$\gamma_P(a_i) + \eta_P(a_i)$, $\gamma_P(a_i) + \theta_P(a_i)$, $\gamma_P(a_i) + \xi_P(a_i)$, $\eta_P(a_i) + \theta_P(a_i)$, $\eta_P(a_i) + \xi_P(a_i)$ and $\theta_P(a_i) + \xi_P(a_i)$ cannot be negative.

Therefore, we deduce that each term in the summation should be equal to 0.

$$\gamma_P(a_i) + \eta_P(a_i)\ln\gamma_P(a_i) + \eta_P(a_i) = 0 \tag{16}$$

$$\Rightarrow \gamma_P(a_i)\ln\gamma_P(a_i) + \eta_P(a_i) + \eta_P(a_i)\ln\gamma_P(a_i) + \eta_P(a_i) = 0$$

Again,

$$\Rightarrow \gamma_P(a_i)\ln\gamma_P(a_i) + \eta_P(a_i) = 0 \text{ or } \eta_P(a_i)\ln\gamma_P(a_i) + \eta_P(a_i) = 0$$

If $\ln\gamma_P(a_i) + \eta_P(a_i) < 0 \Rightarrow \gamma_P(a_i) = 0$

or if $\ln\gamma_P(a_i) + \eta_P(a_i) = 0 \Rightarrow \gamma_P(a_i) \neq 0$ and $\gamma_P(a_i) + \eta_P(a_i) = 1$

Similarly, If $\ln\gamma_P(a_i) + \eta_P(a_i) < 0 \Rightarrow \eta_P(a_i) = 0$

or if $\ln\gamma_P(a_i) + \eta_P(a_i) = 0 \Rightarrow \eta_P(a_i) \neq 0$ and $\gamma_P(a_i) + \eta_P(a_i) = 1$

The same steps can be applied to other degrees as well.

From the above results we can conclude that,

$$\gamma_P(a_i) = \theta_P(a_i) = \eta_P(a_i) = \xi_P(a_i) = 0 \tag{17}$$

or $\gamma_P(a_i) = \theta_P(a_i) = \eta_P(a_i) = \xi_P(a_i) \neq 0$ when $\gamma_P(a_i) + \eta_P(a_i) = \gamma_P(a_i) + \theta_P(a_i) = \gamma_P(a_i) + \xi_P(a_i) = \eta_P(a_i) + \theta_P(a_i) = \eta_P(a_i) + \xi_P(a_i) = \theta_P(a_i) + \xi_P(a_i) = 1$.

But, if $\gamma_P(a_i) + \eta_P(a_i) = 1$, then $\xi_P(a_i)$ and $\theta_P(a_i)$ must be equal to 0 (according to the definition) and from $\gamma_P(a_i) + \xi_P(a_i) = 1 \Rightarrow \gamma_P(a_i) = 1$.

Therefore, we can conclude that,

$$\begin{aligned} &\gamma_P(a_i) = 1 \text{ and } \eta_P(a_i) = \theta_P(a_i) = \xi_P(a_i) = 0 \\ &\eta_P(a_i) = 1 \text{ and } \gamma_P(a_i) = \theta_P(a_i) = \xi_P(a_i) = 0 \\ &\theta_P(a_i) = 1 \text{ and } \gamma_P(a_i) = \eta_P(a_i) = \xi_P(a_i) = 0 \\ &\xi_P(a_i) = 1 \text{ and } \gamma_P(a_i) = \eta_P(a_i) = \theta_P(a_i) = 0 \end{aligned} \tag{18}$$

P is a crisp set.

PIFS2. Maximality

$$Let \gamma_P(a_i) = \eta_P(a_i) = \theta_P(a_i) = \xi_P(a_i) = \frac{1}{4}, \forall a_i \epsilon U \tag{19}$$

$$\Rightarrow E_{PIFS} = \frac{-1}{3n\ln 2}\sum_{i=1}^{n}\frac{1}{2}\ln\frac{1}{2} + \frac{1}{2}\ln\frac{1}{2} + \frac{1}{2}\ln\frac{1}{2} + \frac{1}{2}\ln\frac{1}{2}$$

$$\Rightarrow E_{PIFS} = \frac{-1}{3n\ln 2}\sum_{i=1}^{n} 6\times\frac{1}{2}\ln\frac{1}{2}$$

$$\Rightarrow E_{PIFS} = \frac{-1}{3n\ln 2}\times 3n\ln\frac{1}{2}$$

$$\Rightarrow E_{PIFS} = 1 \tag{20}$$

Now,

$$\text{We have } \gamma_P\ a_i\ +\theta_P\ a_i\ +\eta_P\ a_i\ +\xi_P\ a_i\ =1, \tag{21}$$

$$\text{Therefore, } l\ \gamma_P,\theta_P,\eta_P,\xi_P\ =\gamma_P\ a_i\ +\theta_P\ a_i\ +\eta_P\ a_i\ +\xi_P\ a_i\ -1 \tag{22}$$

In order to get the maximum value of PIFS entropy $E_{PIFS}(P)$ we will construct a Lagrange function $L\ \gamma_P,\theta_P,\eta_P,\xi_P$ with Lagrange multiplier as λ.

$$L\ \gamma_P,\theta_P,\eta_P,\xi_P\ =E_{PIFS}\ \gamma_P,\theta_P,\eta_P,\xi_P\ +\lambda l\ \gamma_P,\theta_P,\eta_P,\xi_P \tag{23}$$

$$\begin{aligned}L=\frac{-1}{3n\ln 2}\sum_{i=1}^{n}&\ \gamma_P\ a_i\ +\eta_P\ a_i\ \ \ln\ \gamma_P\ a_i\ +\eta_P\ a_i\\&+\ \gamma_P\ a_i\ +\theta_P\ a_i\ \ \ln\ \gamma_P\ a_i\ +\theta_P\ a_i\ \ +\ \gamma_P\ a_i\ +\xi_P\ a_i\\&\times\ln\ \gamma_P\ a_i\ +\xi_P\ a_i\ \ +\ \eta_P\ a_i\ +\theta_P\ a_i\ \ \ln\ \eta_P\ a_i\ +\theta_P\ a_i\\&+\ \eta_P\ a_i\ +\xi_P\ a_i\ \ \ln\ \eta_P\ a_i\ +\xi_P\ a_i\ \ +\ \theta_P\ a_i\ +\xi_P\ a_i\\&\times\ln\ \theta_P\ a_i\ +\xi_P\ a_i\ \ +\lambda\gamma_P\ a_i\ +\lambda\eta_P\ a_i\\&+\lambda\theta_P\ a_i\ +\lambda\xi_P\ a_i\ -\lambda\end{aligned} \tag{24}$$

Partially differentiating by $\partial\gamma_p$, we get

$$\begin{aligned}&\frac{\partial k}{\partial\gamma_P}=1+\ln\ \gamma_P+\eta_P\ \ +1+\ln\ \gamma_P+\theta_P\ \ +1+\ln\ \gamma_P+\xi_P\ \ +\lambda=0\\&\frac{\partial k}{\partial\gamma_P}=\ln\ \gamma_P+\eta_P\ \ +\ln\ \gamma_P+\theta_P\ \ +\ln\ \gamma_P+\xi_P\ \ =-\lambda-3\\&\frac{\partial k}{\partial\gamma_P}=\ln\ \gamma_P+\eta_P\ \ \gamma_P+\theta_P\ \ \gamma_P+\xi_P\ \ =-\lambda-3\end{aligned}$$

$$\frac{\partial k}{\partial\gamma_P}=\ \gamma_P+\eta_P\ \ \gamma_P+\theta_P\ \ \gamma_P+\xi_P\ \ =e^{-\lambda-3} \tag{25}$$

Similarly,

$$\frac{\partial k}{\partial\eta_p}=\ \gamma_P+\eta_P\ \ \eta_P+\theta_P\ \ \eta_P+\xi_P\ \ =e^{-\lambda-3} \tag{26}$$

$$\frac{\partial k}{\partial\theta_p}=\ \theta_P+\gamma_P\ \ \theta_P+\eta_P\ \ \theta_P+\xi_P\ \ =e^{-\lambda-3} \tag{27}$$

$$\frac{\partial k}{\partial\xi_p}=\ \xi_P+\gamma_P\ \ \xi_P+\eta_P\ \ \xi_P+\theta_P\ \ =e^{-\lambda-3} \tag{28}$$

Subtracting Eq. (26) from 25

$$\begin{aligned}&\gamma_P+\eta_P\ \ {\gamma_P}^2+\gamma_P\xi_P+\gamma_P\theta_P+\theta_P\xi_P-{\eta_P}^2-\eta_P\xi_P-\eta_P\theta_P-\theta_P\xi_P\ =0\\&\gamma_P+\eta_P\ \ {\gamma_P}^2-{\eta_P}^2+\gamma_P\xi_P+\gamma_P\theta_P-\eta_P\xi_P-\eta_P\theta_P\ =0\\&\gamma_P+\eta_P\ \ (\gamma_P+\eta_P)(\gamma_P-\eta_P)+\gamma_P\xi_P-\eta_P\xi_P+\gamma_P\theta_P-\eta_P\theta_P\ =0\\&\gamma_P+\eta_P\ \ (\gamma_P+\eta_P)(\gamma_P-\eta_P)+\xi_P(\gamma_P-\eta_P)+\theta_P(\gamma_P-\eta_P)\ =0\\&\gamma_P+\eta_P\ \ \gamma_P-\eta_P\ \ \gamma_P+\eta_P+\xi_P+\theta_P\ =0\\&\gamma_P+\eta_P\ \ \gamma_P-\eta_P\ =0\\&{\gamma_P}^2-{\eta_P}^2\ =0\\&\gamma_P=\eta_P\end{aligned} \tag{29}$$

Similarly, by subtracting Eq. (28) from (27), (27) from (25) and (28) from (26), we get

$$\begin{aligned}&\theta_P=\xi_P\\&\gamma_P=\theta_P\\&\eta_P=\xi_P\end{aligned} \tag{30}$$

$$\text{Since, } \gamma_P+\eta_P+\xi_P+\theta_P=1 \tag{31}$$

$$\text{Therefore, } \gamma_P=\eta_P=\xi_P=\theta_P=\frac{1}{4} \tag{32}$$

PIFS3. Resolution

After generating the hessian matrix for $E_{PIFS}(P)$, we get,

$$H\ \ E_{PIFS}(P)\ =\frac{3}{n\ln(2)}\begin{bmatrix}-1&0&0&0\\0&-1&0&0\\0&0&-1&0\\0&0&0&-1\end{bmatrix} \tag{33}$$

It is coming out to be negative definite and hence it is strictly concave with maximum stationary point as $\gamma_P\ a_i\ =\theta_P\ a_i\ =\eta_P\ a_i\ =\xi_P\ a_i\ =\frac{1}{4}$.

The function $E_{PIFS}(P)$ is strictly concave with respect to $P\in PIFS(U)$, and it attains its maximum value at a stationary point. If the maximum value of $\gamma_P\ a_i\ ,\theta_P\ a_i\ ,\eta_P\ a_i\ $ is less than or equal to 1/4, then it implies that $\gamma_P\ a_i\ \le\gamma_Q\ a_i\ ,\theta_P\ a_i\ \le\theta_Q\ a_i\ ,\eta_P\ a_i\ \le\eta_Q\ a_i\ $, which further implies that $\xi_P\ a_i\ \ge\xi_Q\ a_i\ \ge 1/4$. Therefore, from the above discussion it is safe to conclude that PIFS3 condition is fulfilled by $E_{PIFS}(P)$.

On the other hand, if the minimum value of $\{\gamma_P\ a_i\ ,\theta_P\ a_i\ ,\eta_P\ a_i\ \}$ is greater than or equal to 1/4, then it implies that $\gamma_P\ a_i\ \ge\gamma_Q\ a_i\ ,\theta_P\ a_i\ \ge\theta_Q\ a_i\ ,\eta_P\ a_i\ \ge\eta_Q\ a_i\ $. Again, using property the same property, we can conclude that $E_{PIFS}(P)$ satisfies condition PIFS3.

PIFS4. Symmetry

$$\begin{aligned}E_{PIFS}(P^C)=\frac{-1}{3n\ln 2}\sum_{i=1}^{n}&\ \gamma_P\ a_i\ +\eta_P\ a_i\ \ \ln\ \gamma_P\ a_i\ +\eta_P\ a_i\\&+\ \gamma_P\ a_i\ +\theta_P\ a_i\ \ \ln\ \gamma_P\ a_i\ +\theta_P\ a_i\\&+\ \gamma_P\ a_i\ +\xi_P\ a_i\ \ \ln\ \gamma_P\ a_i\ +\xi_P\ a_i\\&+\ \eta_P\ a_i\ +\theta_P\ a_i\ \ \ln\ \eta_P\ a_i\ +\theta_P\ a_i\\&+\ \eta_P\ a_i\ +\xi_P\ a_i\ \ \ln\ \eta_P\ a_i\ +\xi_P\ a_i\\&+\ \theta_P\ a_i\ +\xi_P\ a_i\ \ \ln\ \theta_P\ a_i\ +\xi_P\ a_i\end{aligned} \tag{34}$$

$$E_{PIFS}(P^C)=E_{PIFS}(P)$$

It becomes evident that our proposed entropy measure for PIFS fulfills all the axioms established for entropy measures of PIFS. This finding provides strong evidence of the measure's validity and its ability to accurately capture the uncertainty inherent in the data. Therefore, our proposed measure can be considered a reliable and robust tool for quantifying entropy in PIFS.

4. Validation of the entropy

Example 1. Let $X=\{x_1,x_2,x_3,x_4,x_5\}$ be a universal set.

In the characterization of linguistic variables, we regard

A as "Large"

A^2 as "Very Large"

A^3 as "Quite Very Large"

A^4 as "Very Very Large"

According to the following operational law (where n is a real number), we will calculate the values of A, A^2, A^3, and A^4

$$\beta_1^n=\ \ \gamma\beta_1+\theta\beta_1\ ^n-\theta\beta_1^n,\theta\beta_1^n,1-\ \ 1-\eta\beta_1\ ^n\ \ ,n>0$$

Based on the mathematical view, the entropy measure E must satisfy -

$$E(A)>E(A^2)>E(A^3)>E(A^4)$$

After calculating the entropy of each set using the proposed entropy method, we get the following results: -

Linguistic variables	ENTROPY
A	0.8135
A^2	0.7853
A^3	0.7571
A^4	0.7192

Table 1
Data of linguistic variables.

	x_1			x_2			x_3			x_4			x_5		
	γ	θ	η	γ	θ	η	γ	θ	η	γ	θ	η	γ	θ	η
A	0.100	0.400	0.400	0.300	0.500	0.200	0.600	0.200	0.100	0.800	0.100	0.100	0.500	0.200	0.100
A^2	0.090	0.160	0.640	0.390	0.250	0.360	0.600	0.040	0.190	0.800	0.010	0.190	0.450	0.040	0.190
A^3	0.061	0.064	0.784	0.387	0.125	0.488	0.504	0.008	0.271	0.728	0.001	0.271	0.335	0.008	0.271
A^4	0.037	0.026	0.870	0.347	0.063	0.590	0.408	0.002	0.344	0.656	0.000	0.344	0.239	0.002	0.344

Fig. 1. Methodology's Framework.

Table 1 establishes the entropy-based conditions for entropy measure. Hence, validated.

5. Proposed entropy based TOPSIS for MCDM

In this segment, we have explained the methodology of solving an MCDM problem by using TOPSIS technique under Picture Fuzzy environment. The criteria's weights are computed with the help of the proposed entropy method which is of utmost importance in decision making process. The entire methodology consists of 8 steps and each step is explained in detail. The flow of the methodology is illustrated in Fig. 1

Step 1: Determination of objectives, alternative and criteria

In this step, the DM defines the objective of the problem and then identifies the criteria and the attributes to consider for the evaluation.

Step 2: Formulation of Decision Matrix

The DM will give its opinion on the basis of its knowledge and expertise for each alternative with respect to each criterion and that will be recorded in PIFN. Therefore, a decision matrix will be formulated.

Step 3: Normalization of the Decision Matrix.

The normalization of the matrix is done according to the pre-defined methods.

Step 4: Determination of weights of criteria using proposed entropy method.

$$E_{PIFS}(a_j) = \frac{-1}{3m\ln 2}\sum_{i=1}^{m} \gamma_{ij}+\eta_{ij}\ \ln\ \gamma_{ij}+\eta_{ij}\ +\ \gamma_{ij}+\theta_{ij} \times \ln\ \gamma_{ij}+\theta_{ij}\ +\ \gamma_{ij}+\xi_{ij}\ \ln\ \gamma_{ij}+\xi_{ij} +\ \eta_{ij}+\theta_{ij}\ \ln\ \eta_{ij}+\theta_{ij}\ +\ \eta_{ij}+\xi_{ij}\ \ln\ \eta_{ij}+\xi_{ij} +\ \theta_{ij}+\xi_{ij}\ \ln\ \theta_{ij}+\xi_{ij} \tag{35}$$

Where a_j denotes the jth criterion ($i = 1, 2, 3, \ldots, m$ *and* $j = 1, 2, 3, \ldots, n$)

$$d_j^P = 1 - E_{PIFS}(a_j) \tag{36}$$

d_j^P denotes the degree of divergence which tends to increase with decrease in entropy value. Therefore, formula for weight is given below.

$$w_j^P = \frac{d_j^P}{\sum_{j=1}^{n} d_j^P} \tag{37}$$

w_j^P denotes the weight of the criterion. In conclusion, lesser entropy value indicates, greater weight of the criterion,

Step 5: Formation of a weighted aggregated Decision Matrix

After the calculation of weights of criteria, an aggregated decision matrix is formed with the help of scalar multiplication of weights, as follows

$$w_j^P a_{ij} = \ 1 - \ 1-\gamma_{ij}\ ^{w_j^P}, \theta_{ij}^{\ w_j^P}, \eta_{ij}^{\ w_j^P}\ \ , \text{ where } w_j^P > 0. \tag{38}$$

Step 6: Calculation of PIF +ve ideal solution $^+$ and PIF -ve ideal solution $^-$

From this step, we will start the procedure of TOPSIS for which we need to calculate PIF +ve ideal solution, denoted by ρ^+ and PIF -ve ideal solution denoted by ρ^-.

$$\rho^+ = \ \gamma_j^+, \theta_j^+, \eta_j^+ \ = \begin{cases} \langle \max_j \gamma_{ij}, \min_j \theta_{ij}, \min_j \eta_{ij} \rangle & : C_j \epsilon C_B \\ \langle \max_j \gamma_{ij}, \max_j \theta_{ij}, \min_j \eta_{ij} \rangle & : C_j \epsilon C_{NB} \end{cases} \tag{39}$$

$$\rho^- = \ \gamma_j^-, \theta_j^-, \eta_j^- \ = \begin{cases} \langle \min_j \gamma_{ij}, \min_j \theta_{ij}, \max_j \eta_{ij} \rangle & : C_j \epsilon C_B \\ \langle \min_j \gamma_{ij}, \max_j \theta_{ij}, \max_j \eta_{ij} \rangle & : C_j \epsilon C_{NB} \end{cases} \tag{40}$$

Step 7: Calculation of the degree of weighted similarity

We will calculate the degree of weighted similarity S^+ between ρ^+ and each alternative. Also, we will evaluate the degree of weighted